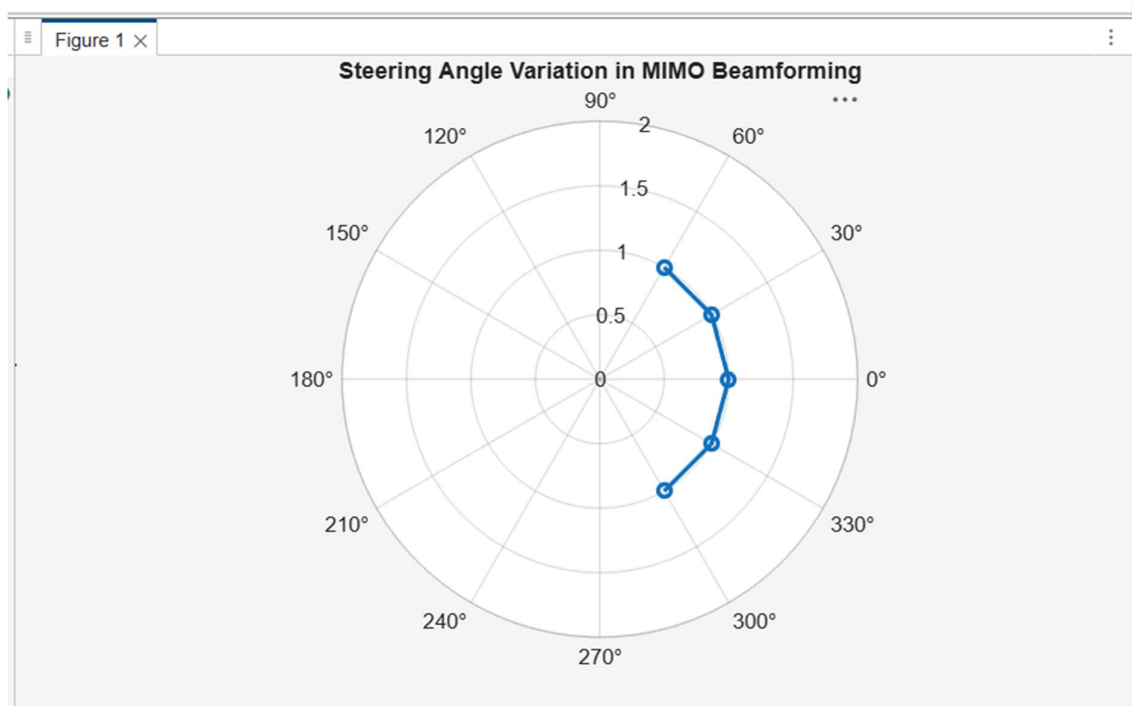


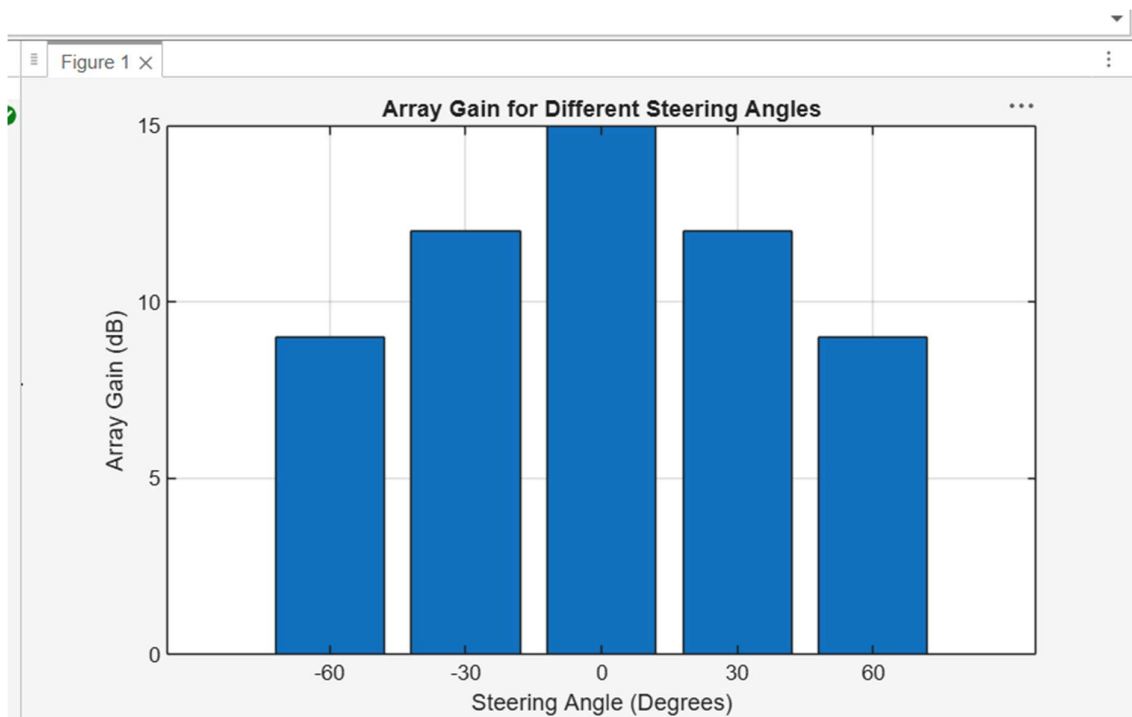
Objective 1

```
clc;  
clear;  
close all;  
angle = [-60 -30 0 30 60];  
gain = ones(size(angle));  
figure  
% Steering Angles (degrees)  
% Normalized Gain  
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)  
title('Steering Angle Variation in MIMO Beamforming')
```



Objective 2

```
clc;  
clear;  
close all;  
angle = [-60 -30 0 30 60];  
gain = [9 12 15 12 9];  
figure  
bar(angle, gain)  
grid on  
% Steering Angles (degrees)  
% Array Gain (dB)  
xlabel('Steering Angle (Degrees)')  
ylabel('Array Gain (dB)')  
title('Array Gain for Different Steering Angles')
```



Objective 3

```
clc;
```

```
clear;
```

```
close all;
```

```
desired = [-60 -30 0 30 60];
```

```
actual = [-58 -32 1 29 62];
```

```
error = abs(desired - actual);
```

```
figure
```

```
% Desired Steering Angle (degrees)
```

```
% Actual Beam Direction (degrees)
```

```
% Beam Direction Error
```

```
plot(desired, error, 'o-', 'LineWidth', 2)
```

```
grid on
```

```
xlabel('Desired Steering Angle (Degrees)')
```

```
ylabel('Beam Direction Error (Degrees)')
```

```
title('Beam Direction Error for Different Steering Angles')
```

